

Total Marks – 120
Attempt Questions 1-8
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet. **Marks**

(a) Use Integration by parts to find $\int x e^{-3x} dx$ **2**

(b) Use the substitution $x = \frac{2}{3} \sin \alpha$ to prove that $\int_0^{\frac{2}{3}} \sqrt{4 - 9x^2} dx = \frac{\pi}{3}$ **3**

(c) Use the table of standard integrals to help evaluate $\int \frac{dx}{\sqrt{x^2 - 4x + 29}}$ **2**

(d) Evaluate that $\int_4^6 \frac{2dt}{(t-1)(t-3)}$ **4**

(e) Use the substitution $t = \tan \frac{x}{2}$ to show that **4**

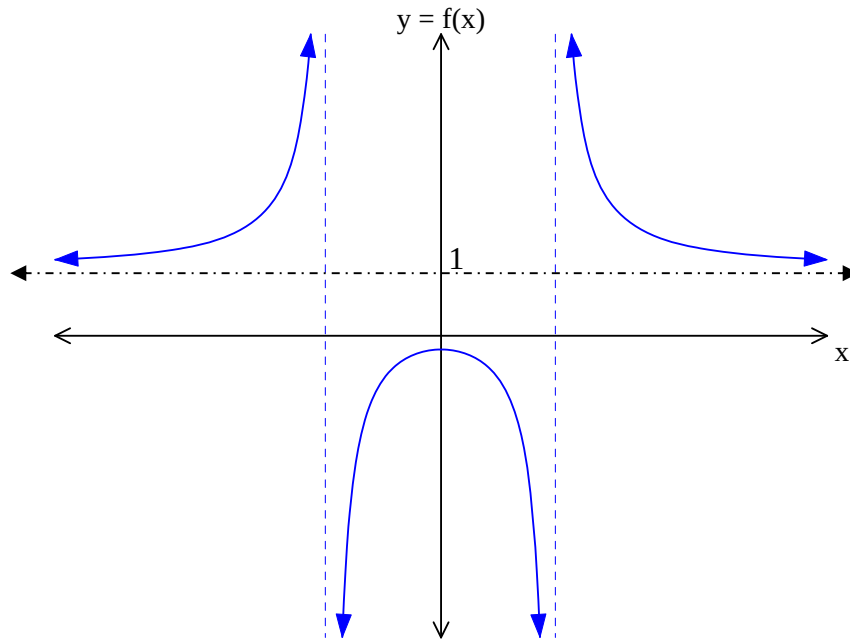
$$\int_0^{\frac{\pi}{2}} \frac{dx}{5 + 4 \cos x} = \frac{2}{3} \tan^{-1} \frac{1}{3}$$

Question 2 (15 marks) Use a SEPARATE writing booklet.**Marks**

- (a) For the complex number $z = 1 - \sqrt{3}i$, express each of the following in the form $a + bi$, (a, b are real numbers). 3
- (i) $\bar{z}$
 (ii) z^2
 (iii) $\frac{1}{z}$
- (b) If $z_1 = 1 - 2i$; $z_2 = 2 + i$ and $z = \frac{z_1}{z_2}$ find :- 3
- i) $|z|$
 ii) $\arg(z)$
- (c) Prove that $|z|^2 = z\bar{z}$ for all complex numbers z . 2
- (d) If ω is a complex cube root of unity,
- (i) Write down the value of $1 + \omega + \omega^2$. 1
- (ii) Simplify $\omega^4 + \omega^5 + \omega^6$. 1
- (e) Sketch on an Argand diagram the region in which z lies, showing all important features where $2 \leq |z| \leq 3$ and $\frac{\pi}{4} < \arg(z - i) < \frac{3\pi}{4}$. 2
- (f) P_1 and P_2 are points representing the complex numbers z_1 and z_2 on an Argand diagram. If OP_1P_2 is an isosceles triangle and angle P_1OP_2 is a right-angle, show that $z_1^2 + z_2^2 = 0$. 3

Question 3 (15 marks) Use a SEPARATE writing booklet.**Marks**

- (a) The diagram below shows the graph of the function $y = f(x)$ where $f(x) = \frac{1+x^2}{x^2-9}$.



Draw a separate sketch of each of the following graphs.

Use about one third of a page for each graph. Show all significant features.

- | | |
|--|---|
| (i) $y = [f(x)]^2$ | 2 |
| (ii) $y = \frac{1}{f(x)}$ | 2 |
| (iii) $y = f'(x)$ | 2 |
| (iv) Draw $y = f(x)$ and $y = \sqrt{f(x)}$ on the same number plane. | 2 |
| (v) $ y = f(x)$ | 2 |
- (b) For the curve defined by $3x^2 + y^2 - 2xy - 8x + 2 = 0$,
- | | |
|--|---|
| (i) Show that $\frac{dy}{dx} = \frac{3x - y - 4}{x - y}$. | 2 |
| (ii) Find the coordinates of the points on the curve where the tangent to the curve is parallel to the line $y = 2x$. | 3 |

Question 4 (15 marks) Use a SEPARATE writing booklet.**Marks**

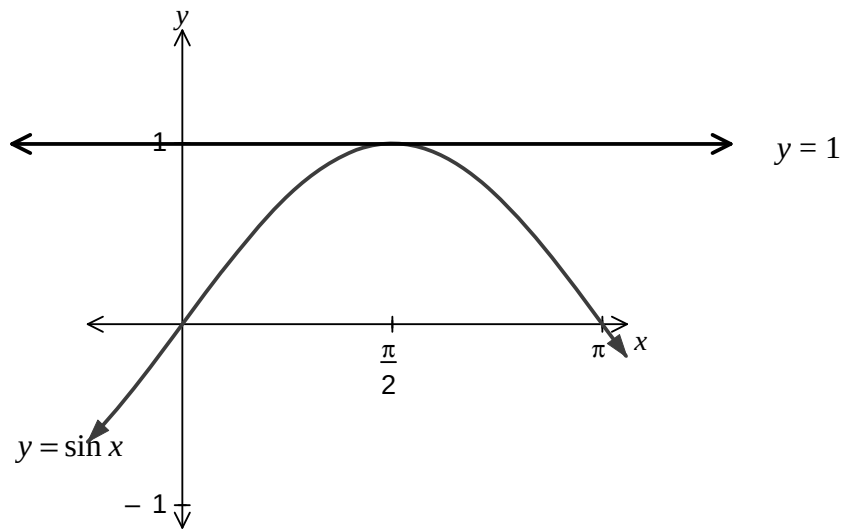
- (a) The equation $x^3 + 2x^2 + bx - 16 = 0$ has roots α, β and γ such that $\alpha\beta = 4$.
- (i) Show that $b = -20$. 2
- (ii) Find the equation with roots given by α^2, β^2 and γ^2 2
- (b) Consider the polynomial $P(x) = (x - \alpha)^3 \cdot Q(x)$, where $Q(x)$ is also a polynomial and α is a real zero of $P(x)$.
- (i) Show that $P(\alpha) = P'(\alpha) = P''(\alpha) = 0$ 2
- (ii) Hence or otherwise, solve the equation $8x^4 - 25x^3 + 27x^2 - 11x + 1 = 0$ given that it has a triple root. 2
- (c) (i) Show that the solutions of $z^6 + z^3 + 1 = 0$ are contained in the solutions of $z^9 - 1 = 0$. 2
- (ii) Sketch the nine solutions of $z^9 - 1 = 0$ on an Argand Diagram. (about one third of a page in size) 2
- (iii) Mark clearly on your diagram, the six roots $z_1, z_2, z_3, z_4, z_5, z_6$ of $z^6 + z^3 + 1 = 0$. 4
- (iv) Show that the sum of the six roots of $z^6 + z^3 + 1 = 0$ can be given by $2\left(\cos\frac{2\pi}{9} + \cos\frac{4\pi}{9} - \cos\frac{\pi}{9}\right)$ 2

Question 5 (15 marks) Use a SEPARATE writing booklet.**Marks**

- (a) For the curve with Cartesian equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$, 5
- (i) find the eccentricity and the coordinates of a focus
(ii) find the equation of the corresponding directrix
(iii) hence write down the coordinates of a focus and the equation of the corresponding directrix for the curve with the Cartesian equation $\frac{x^2}{4} + \frac{y^2}{9} = 1$.
- (b) Let $P(4\sec\theta, 3\tan\theta)$ be any point on the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$. 4
- (i) Derive the equations of the tangent and normal at P and show that they are respectively:
- $$3x\sec\theta - 4y\tan\theta = 12 \text{ and } 4x\tan\theta + 3y\sec\theta = 25\sec\theta\tan\theta$$
- (ii) The tangent and normal at P meet the y -axis at T and N respectively. 2
Show that
- $$T = (0, -3\cot\theta) \text{ and } N = (0, \frac{25}{3}\tan\theta).$$
- (iii) Show that the circle with diameter NT passes through a focus. 4

Question 6 (15 marks) Use a SEPARATE writing booklet.**Marks**

(a)



The area defined by $y \geq \sin x$, $0 \leq x \leq \frac{\pi}{2}$ and $0 \leq y \leq 1$ is rotated about the straight line $y = 1$.

- (i) Copy the diagram above into your writing booklet and shade in the region defined by the simultaneous inequalities $y \geq \sin x$, $0 \leq x \leq \frac{\pi}{2}$ and $0 \leq y \leq 1$. **1**
- (ii) Find the total volume of the solid formed, by taking slices perpendicular to the axis of rotation. **3**

- (b) The horizontal base of a solid is an ellipse defined by the equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. **5**

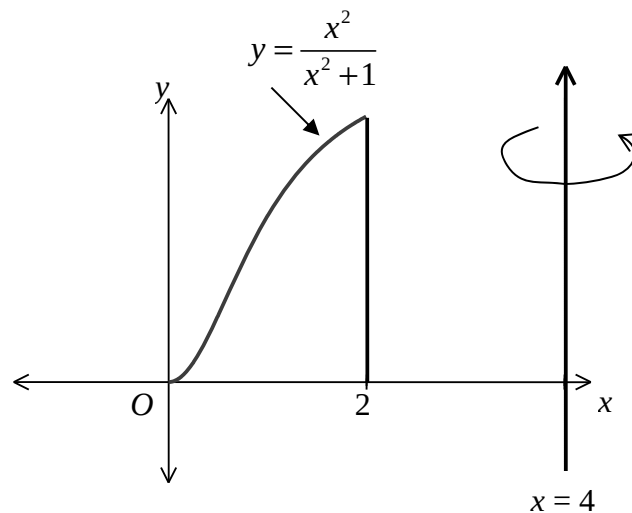
Vertical cross-sections taken perpendicular to the y axis are squares with one side in the horizontal base of the solid.

Find the volume of the solid formed in terms of a and b .

Question 6 continues on the next page ...

Question 6 continued ...**Marks**

- (c) The region bounded by the curve $y = \frac{x^2}{x^2 + 1}$, the x axis and $0 \leq x \leq 2$, is rotated about the line $x = 4$ to form a solid.



- (i) Using the method of cylindrical shells, explain why the volume δV of a typical shell distant x units from the origin and with thickness δx is given by **3**

$$\delta V = 2\pi(4 - x) \left(1 - \frac{1}{1 + x^2} \right) \delta x.$$

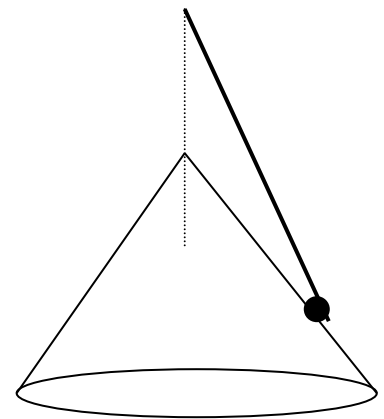
- (ii) Hence, find the total volume of the solid formed. **3**

Question 7 (15 marks) Use a SEPARATE writing booklet.**Marks**

- (a) A particle of mass m is set in motion with speed u . Subsequently the only force acting upon the particle directly opposes its motion and is of magnitude $mk(1 + v^2)$ where k is a constant and v is its speed at time t . 5

- (i) Show that the particle is brought to rest after a time $\frac{1}{k} \tan^{-1} u$.
- (ii) Find an expression for the distance travelled by the particle in this time.

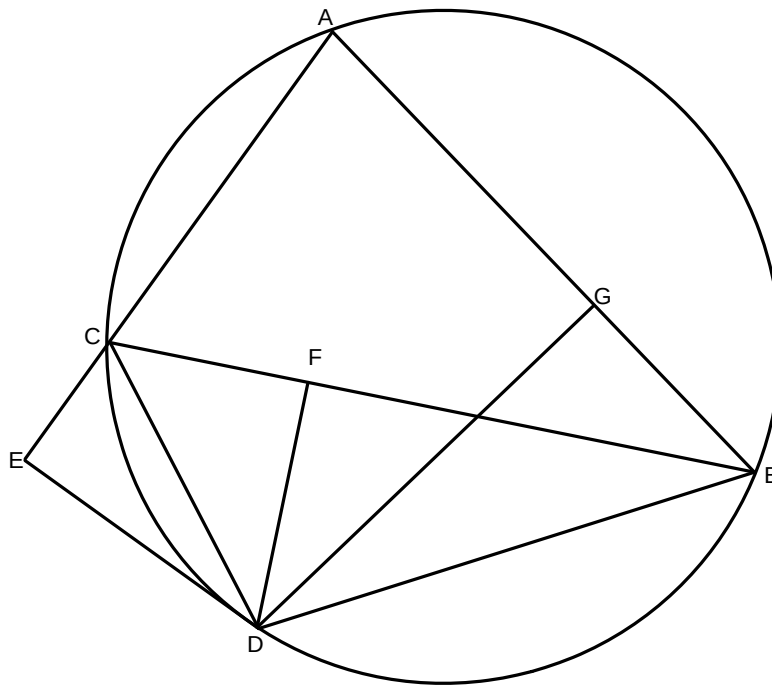
- (b) A smooth circular cone, with its vertex up, and its axis vertical, has a semi-vertex angle of 60° . A particle of mass 1 kg is attached by a light inelastic string from a point vertically above the vertex of the cone, and moves with constant speed $v \text{ m/s}$ on the outer surface of the cone in a horizontal circle of radius 0.5 m . The string makes an angle of 30° with the vertical. Let the magnitude of the tension in the string be T newton, and let the magnitude of the reaction of the cone on the particle be R newton.



- (i) Draw a diagram showing the forces acting on the particle, and the magnitude of the angles made by these forces with the vertical. 2
- (ii) By resolving forces in two directions write down equations of motion for the particle. 2
- (iii) If $v = 1$, find the value of T and R correct to two decimal places. 2
- (iv) Find the maximum value of v in order that the particle remains in contact with the cone. 2
- (v) Deduce the maximum value that T can take if the particle is to remain in contact with the cone. 2

Question 8 (15 marks) Use a SEPARATE writing booklet.**Marks**

(a)



The above diagram shows a triangle ABC inscribed in a circle with D a point on the arc BC . DE is perpendicular to AC produced, DF is perpendicular to BC and DG is perpendicular to AB .

Copy or trace this diagram into your writing booklet.

- | | | |
|------|--|---|
| (i) | Explain why $DECF$ and $DFGB$ are cyclic quadrilaterals. | 2 |
| (ii) | Show that the points E , F and G are collinear. | 3 |

Question 8 continues on the next page...

Question 8 continued...**Marks**

(b) (i) Evaluate $\int_0^{\frac{\pi}{4}} \tan x \, dx$. 2

(ii) If $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$, $n = 0, 1, 2, 3, \dots$, show that for $n = 2, 3, 4, \dots$ 3

$$I_n = \frac{1}{n-1} - I_{n-2}$$

(iii) Hence, evaluate I_5 . 2

(c) If $P(x) = x^m(b^n - c^n) + b^m(c^n - x^n) + c^m(x^n - b^n)$ where m and n are positive integers, show that $x^2 - (b+c)x + bc$ is a factor of $P(x)$. 3

End of paper

EXT 2 TRIAL HSC 2005

Q1

a) $\int x e^{-3x} dx$

let $u = x$ $v' = e^{-3x}$

$u' = 1$ $v = -\frac{1}{3} e^{-3x}$ ①

$\int u v' dx = uv - \int u' v dx$

12

$= -\frac{1}{3} x e^{-3x} + \frac{1}{3} \int e^{-3x} dx$

$= -\frac{1}{3} x e^{-3x} - \frac{1}{9} e^{-3x} + C$ ①

b) $\int_0^{2/3} \sqrt{4-9x^2} dx$ let $x = \frac{2}{3} \sin \alpha$

$\frac{dx}{d\alpha} = \frac{2}{3} \cos \alpha$

$= \int_0^{\pi/2} \sqrt{4-9 \cdot \frac{4}{9} \sin^2 \alpha} \cdot \frac{2}{3} \cos \alpha d\alpha$

$dx = \frac{2}{3} \cos \alpha d\alpha$ ①

$= \frac{2}{3} \int_0^{\pi/2} 2 \sqrt{\cos^2 \alpha} \cdot \cos \alpha d\alpha$ when $x=0$ $\alpha=0$

$x=2/3$ $\alpha = \pi/2$

$= \frac{4}{3} \int_0^{\pi/2} \cos^2 \alpha d\alpha$

$= \frac{4}{3} \int_0^{\pi/2} \frac{1}{2} (\cos 2\alpha + 1) d\alpha$ ①

13

$= \frac{2}{3} \left[\frac{1}{2} \sin 2\alpha + \alpha \right]_0^{\pi/2}$

$= \frac{\pi}{3}$ ①

$\therefore \int_0^{2/3} \sqrt{4-9x^2} dx = \frac{\pi}{3}$

$$\begin{aligned}
 c) \quad & \int \frac{dx}{\sqrt{x^2 - 4x + 29}} \\
 &= \int \frac{dx}{\sqrt{(x-2)^2 + 25}} \quad (1) \quad 1/2 \\
 &= \ln(x-2 + \sqrt{x^2 - 4x + 29}) \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 d) \quad & \int_4^6 \frac{2dt}{(t-1)(t-3)} \\
 & \frac{2}{(t-1)(t-3)} = \frac{A}{t-1} + \frac{B}{t-3} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 2 &= A(t-3) + B(t-1) \\
 \text{Let } t=1 & \quad t=3 \quad (1) \\
 2 &= -2A \quad 2 = 2B \\
 A &= -1 \quad B = 1
 \end{aligned}$$

$$\begin{aligned}
 \therefore \int_4^6 \frac{2dt}{(t-1)(t-3)} &= \int_4^6 \frac{-1}{t-1} + \frac{1}{t-3} dt \quad (1) \\
 &= \left[\ln\left(\frac{t-3}{t-1}\right) \right]_4^6 \quad (1) \\
 &= \ln \frac{3}{5} - \ln \frac{1}{3} \\
 &= \ln \frac{9}{5} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 e) \quad & \text{when } t = \tan \frac{x}{2}, \quad \cos x = \frac{1-t^2}{1+t^2} \quad (1) \quad \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} \\
 \int_0^{\pi/2} \frac{dx}{5+4\cos x} &= \int_0^1 \frac{1}{5+4\left(\frac{1-t^2}{1+t^2}\right)} \cdot 2 \frac{1}{(1+t^2)} dt = \frac{1}{2} (1 + \tan^2 \frac{x}{2}) \\
 & \quad dt = \frac{1}{2} (1+t^2) dx \quad (1) \\
 & \quad \therefore dx = 2 \frac{1}{1+t^2} dt \\
 &= 2 \int_0^1 \frac{1}{5+5t^2+4-4t^2} dt \quad (1) \\
 &= 2 \int_0^1 \frac{1}{9+t^2} dt \quad (1) \\
 &= \frac{2}{3} \left[\tan^{-1} \frac{t}{3} \right]_0^1 \quad (1) \\
 &= \frac{2}{3} \tan^{-1} \left(\frac{1}{3} \right)
 \end{aligned}$$

Q2

a) i, $\bar{z} = 1 + \sqrt{3}i$ ① ans

ii, $z^2 = (1 - \sqrt{3}i)^2$
 $= 1 - 2\sqrt{3}i + 3i^2$
 $= -2 - 2\sqrt{3}i$ ① ans

iii, $\frac{1}{z} = \frac{1}{1 - \sqrt{3}i} \times \frac{1 + \sqrt{3}i}{1 + \sqrt{3}i}$
 $= \frac{1 + \sqrt{3}i}{1 + 3}$
 $= \frac{1}{4} + \frac{\sqrt{3}}{4}i$ ① ans

b) $|z_1| = \sqrt{5}$ $|z_2| = \sqrt{5}$
 $\arg z_1 = \tan^{-1}(2)$ $\arg z_2 = \tan^{-1}(\frac{1}{2})$

i, $|z| = \frac{|z_1|}{|z_2|}$
 $= 1$ ① ans



ii, $\arg(z) = \arg(z_1) - \arg(z_2)$
 $= \tan^{-1}(2) - \tan^{-1}(\frac{1}{2})$

Let $\alpha = \tan^{-1} 2$ $\beta = \tan^{-1} \frac{1}{2}$

$\therefore 2 = \tan \alpha$

$\tan \beta = \frac{1}{2}$

$\tan(\arg(z)) = \frac{2 - \frac{1}{2}}{1 - 2 \cdot \frac{1}{2}}$

$= \frac{\frac{3}{2}}{0}$

$\therefore \arg z = -\pi/2$ ① ans

① working

c) Let $z = a + ib$ where $z \in \mathbb{C}$ and $a, b \in \mathbb{R}$

LHS

$$|z| = \sqrt{a^2 + b^2}$$

$$\therefore |z|^2 = a^2 + b^2$$

① LHS

RHS

$$z\bar{z} = (a + ib)(a - ib)$$

$$= a^2 + aib - aib - i^2 b^2$$

$$= a^2 + b^2$$

$$\therefore |z|^2 = z\bar{z} \quad \forall z \in \mathbb{C}$$

① RHS

d) i/ $\omega^3 = 1$

$$\omega^3 - 1 = 0$$

$$\therefore (\omega - 1)(1 + \omega + \omega^2) = 0$$

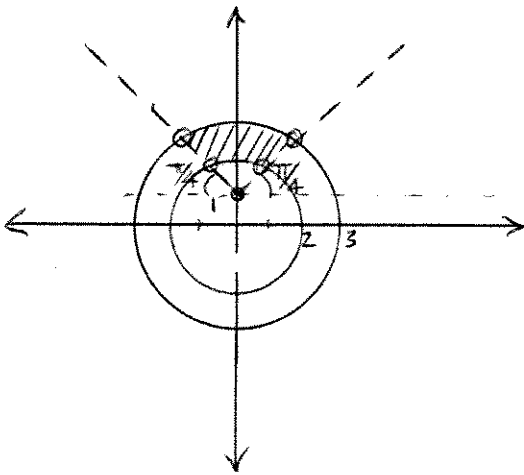
$$\therefore 1 + \omega + \omega^2 = 0 \text{ if } \omega \neq 1. \quad \text{① statement w/ condition.}$$

$$= 3 \text{ if } \omega = 1$$

ii/ $\omega^4 + \omega^5 + \omega^6 = \omega^4(1 + \omega + \omega^2)$

$$= 0 \text{ or } 3\omega? \quad \text{①}$$

e)



① circular region w/ radii + solid lines

① angular region begins at $(0, 1)$

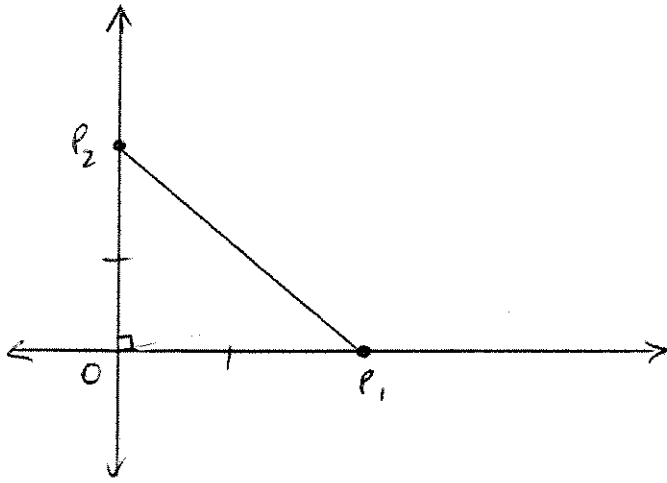
with correct angles + dotted lines

or

① Right idea

② Right idea with detail.

f)



① Diagram

$$|z_1| = |z_2|$$

$$\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2} \therefore z_2 = iz_1$$

$$\therefore |z_1|^2 + |z_2|^2 = |z_1 - z_2|^2$$

Let $z_1 = a$ and $z_2 = ai$

$$\text{Then } z_1^2 + z_2^2 = a^2 - a^2 = 0$$

① specific case

Similarly, in the general case

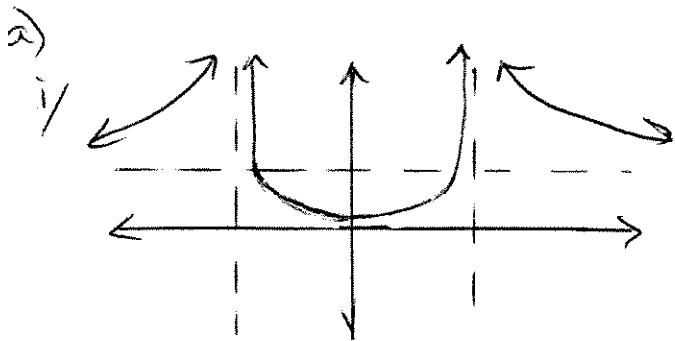
$$z_2 = iz_1$$

$$\therefore z_1^2 + z_2^2 = z_1^2 + (iz_1)^2 \quad \text{① general case}$$

$$= z_1^2 - z_1^2$$

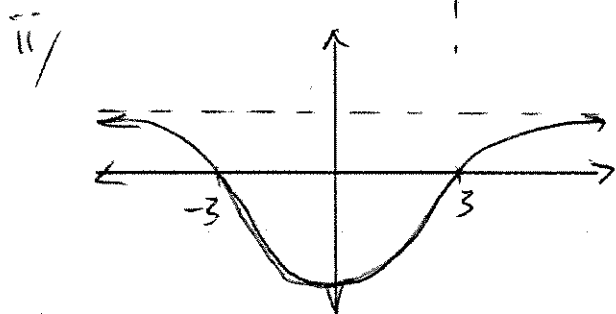
$$= 0$$

Q3



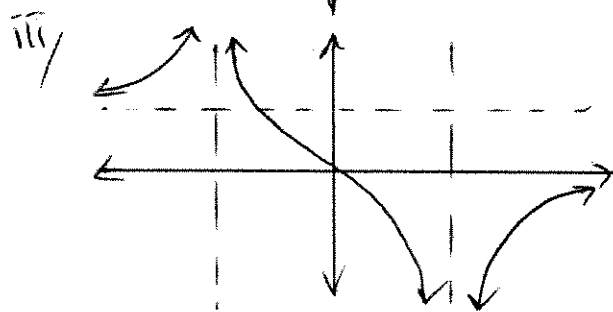
① Branches less steep

① Centre section reflected
(should also be flatter
but no marks deducted
for this)



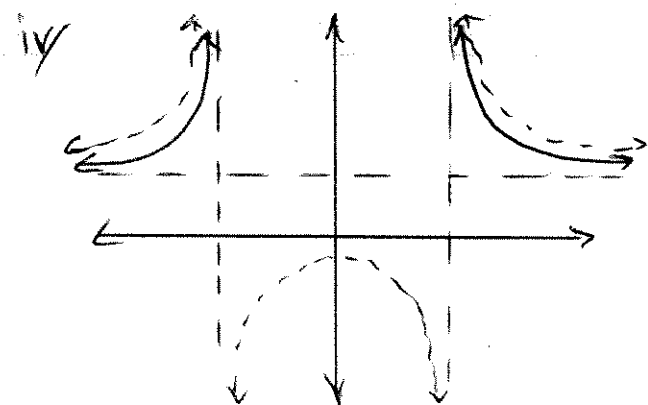
① Intercepts at 3, -3

① Shape



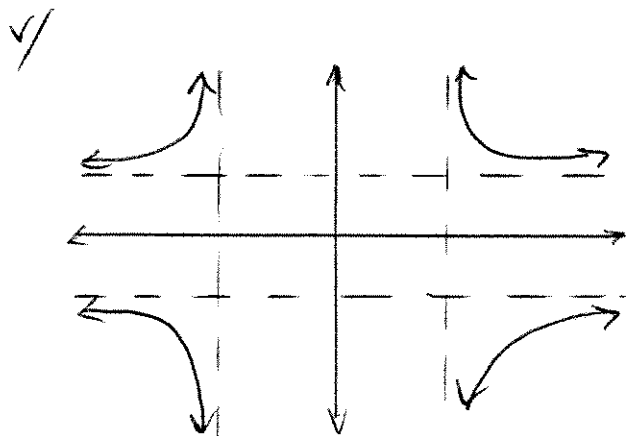
① Signs

① Shape



① Branches below original

① No $y = \sqrt{f(x)}$ for $f(x) < 0$



① Reflection over y of some section

① Correctly reflects only +ve y onto
-y axis

b) i/ $3x^2 + y^2 - 2xy - 8x + 2 = 0$ ① chain rule
 $6x + 2y \frac{dy}{dx} - 2x \frac{dy}{dx} - 2y - 8 = 0$

$$\therefore \frac{dy}{dx} 2(y-x) = -6x + 2y + 8$$

$$\therefore \frac{dy}{dx} = \frac{3x-y-4}{x-y} \quad \text{① otherwise correct.}$$

ii/ $n=2$
 $\therefore \frac{3x-y-4}{x-y} = 2$

$$3x - y - 4 = 2x - 2y$$

$$x + y - 4 = 0$$

$$\therefore y = 4 - x \quad \text{① Condition on } x + y$$

$$\therefore 3x^2 + (4-x)^2 - 2x(4-x) - 8x + 2 = 0$$

$$3x^2 + 16 - 8x + x^2 - 8x + 2x^2 - 8x + 2 = 0$$

$$6x^2 - 24x + 18 = 0$$

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$\therefore x = 1 \text{ or } 3$$

$$y = 3 \text{ or } 1$$

① Solution

① Coordinates

$\therefore$ Coords of pts are $(1, 3) + (3, 1)$.

Q4

a)

i/ $\alpha + \beta = 4$

$\therefore \gamma = -4$

$$b = 4 + \alpha\gamma + \beta\gamma$$

$$= 4 - 4(\alpha + \beta)$$

Now $\alpha + \beta + \gamma = -2$

$\therefore \alpha + \beta = -6$

$\therefore b = 4 - 24$

$\therefore b = -20$

① Reasonable logic

ii/ Let p represent new roots + x the original roots

$p = x^2$

$\therefore x = \sqrt{p}$

$\therefore$ New polynomial has

$$(\sqrt{p})^3 + 2(\sqrt{p})^2 + 20\sqrt{p} - 16 = 0$$

① subst $\sqrt{p}$

$$\sqrt{p}(p - 20) = -2p + 16$$

$$p(p - 20)^2 = (16 - 2p)^2$$

① square + simplify

$$p(p^2 - 40p + 400) = 256 - 64p + 4p^2$$

$$p^3 - 44p^2 + 464p - 256 = 0 \text{ is new polynomial}$$

b) i/ $P(\alpha) = 0$ since $P(x) = (x - \alpha)^3 Q(x)$

$= 0$

$$P'(x) = 3(x - \alpha)^2 Q(x) + Q(x)(x - \alpha)^3$$

$$(x - \alpha)^2 (3Q(x) + (x - \alpha)Q'(x))$$

① $P'(x)$

$\therefore P'(\alpha) = 0$

$$P''(x) = 2(x - \alpha)(3Q(x) + (x - \alpha)Q'(x)) + (x - \alpha)^2 (\dots)$$

$\therefore P''(\alpha) = 0$

① $P''(x)$

$\therefore P(\alpha) = P'(\alpha) = P''(\alpha) = 0$

ii/ $P(x) = 8x^4 - 25x^3 + 27x^2 - 11x + 1$

$P'(x) = 32x^3 - 75x^2 + 54x - 11$

$P''(x) = 96x^2 - 150x + 54$

Let $P''(x) = 0$

Then $96x^2 - 150x + 54 = 0$

$16x^2 - 25x + 9 = 0$

$(16x - 9)(x - 1) = 0$

$x = \frac{9}{16} \text{ or } x = 1$ ① Identify triple root

Check $P(1) = 0$ $P'(1) =$

$\therefore P(x) = (x-1)^3(8x-1)$ ① Determine fourth root

$\therefore x = 1, \frac{1}{8}$

c) i/ $z^9 - 1 = 0$

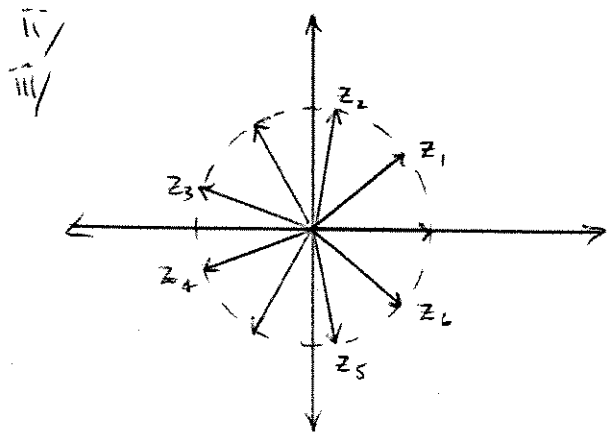
$(z^3 - 1)(z^6 + z^3 + 1) = 0$ ①

Check

$z^9 + z^6 + z^3 - z^6 - z^3 + 1 = 0$ ① Working

$z^9 - 1 = 0$

$\therefore$ Solns of $z^6 + z^3 + 1 = 0$ are contained in $z^9 - 1 = 0$



Roots of $z^3 - 1 = 0$ are $\cos \frac{2\pi}{3}$, $\cos \frac{4\pi}{3}$, 1 .

① Unit circle

① 9 roots marked w/ vectors at equal angles.

① Identify roots of $z^3 - 1$

① Mark all 6 roots of $z^6 + z^3 + 1 = 0$.

iv/ Roots are: $\cos(\pm \frac{2\pi}{9})$, $\cos(\pm \frac{8\pi}{9})$, $\cos(\pm \frac{4\pi}{9})$

① Name roots

$\therefore \text{Sum} = 2\cos \frac{2\pi}{9} + 2\cos \frac{8\pi}{9} + 2\cos \frac{4\pi}{9}$

but $\cos \frac{8\pi}{9} = -\cos \frac{\pi}{9}$

① Identify on cos.

$\therefore \text{Sum} = 2(\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} - \cos \frac{\pi}{9})$

(5) (a) (i) $b^2 = a^2(1 - e^2)$ ①
 $4 = 9(1 - e^2)$ ①
 $e = \frac{\sqrt{5}}{3}$ ①

$S(\sqrt{5}, 0)$ ①

(ii) $x = \frac{9}{\sqrt{5}}$ ①

(iii) $S(0, \sqrt{5}), y = \frac{9}{\sqrt{5}}$ ① both

(b) (i) $\frac{2x}{16} - 2y \frac{dy}{dx} = 0$ ①

$\frac{dy}{dx} = \frac{9x}{16y}$ ①
 $= \frac{3 \sec \theta}{4 + \tan \theta}$

tangent: $y - 3 \tan \theta = \frac{3 \sec \theta}{4 + \tan \theta} (x - 4 \sec \theta)$ ①

normal: $y - 3 \tan \theta = \frac{-4 \tan \theta}{3 \sec \theta} (x - 4 \sec \theta)$ ①

(ii) At T, $x = 0 \therefore y = \frac{-3}{\tan \theta} = -3 \cot \theta$ ① correct intermediate step

At N, $x = 0, \therefore y = \frac{16 \tan \theta}{3} + 3 \tan \theta = \frac{25 \tan \theta}{3}$ ①

(iii) $b^2 = a^2(e^2 - 1)$
 $9 = 16(e^2 - 1)$
 $e = \frac{5}{4}$

$\therefore S = (5, 0)$

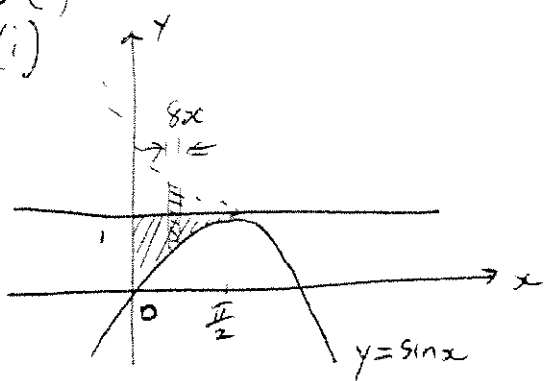
$\therefore$ circle with diameter NT passes through S
 $(\angle NST = \text{angle in semicircle})$

① correct focus
 ② 1 for each gradient

① stating $\angle$ in semicircle

$m_{NS} \cdot m_{TS} = \frac{\frac{25}{3} \tan \theta}{-5} \cdot \frac{-3 \cot \theta}{-5} = -1 \therefore \angle NST = 90^\circ$

5) (a)
(i)



①

(ii)



$$\delta V \doteq \pi r^2 \delta x$$

$$V \doteq \sum_{x=0}^{x=\frac{\pi}{2}} \pi (1-y)^2 \delta x$$

$$= \lim_{\delta x \rightarrow 0} \sum_{x=0}^{x=\frac{\pi}{2}} \pi (1-y)^2 \delta x$$

$$V = \pi \int_0^{\frac{\pi}{2}} (1-y)^2 dx$$

①

$$= \pi \int_0^{\frac{\pi}{2}} (1 - 2y + y^2) dx, \quad y = \sin x$$

$$= \pi \int_0^{\frac{\pi}{2}} \left(1 - 2\sin x + \frac{1}{2}(1 - \cos 2x) \right) dx$$

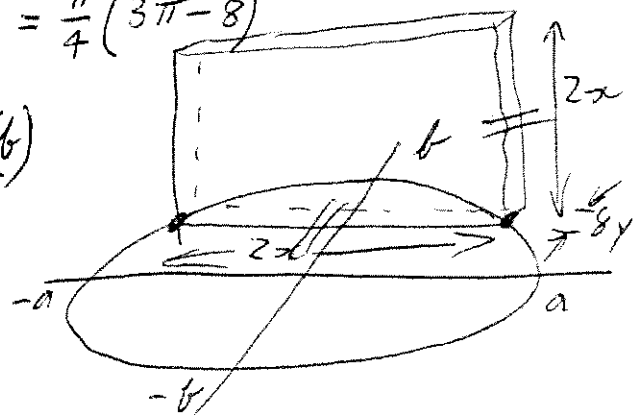
$$= \pi \left[x + 2\cos x + \frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) \right]_0^{\frac{\pi}{2}}$$

①

$$= \frac{\pi}{4} (3\pi - 8)$$

①

(b)



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$x^2 = a^2 \left(1 - \frac{y^2}{b^2} \right)$$

$$\delta V \doteq 4x^2 \delta y$$

$$V \doteq \sum_{y=-b}^{y=b} 4x^2 \delta y$$

$$V = \lim_{\delta y \rightarrow 0} \sum_{y=-b}^{y=b} 4x^2 \delta y$$

6(b) c+d

$$V = \int_{-b}^b 4a^2 \left(1 - \frac{y^2}{b^2}\right) dy$$

$$= 8a^2 \int_0^b \left(1 - \frac{y^2}{b^2}\right) dy$$

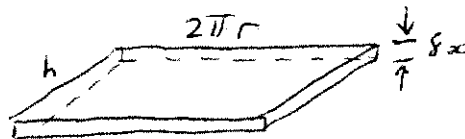
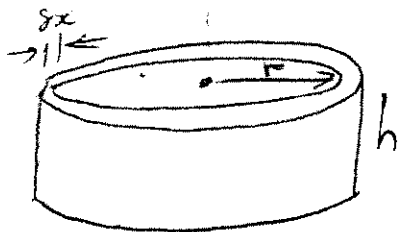
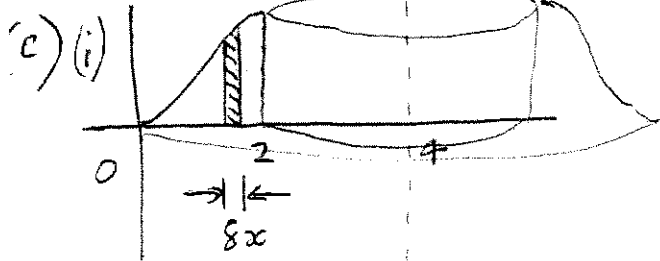
$$= 8a^2 \left[y - \frac{y^3}{3b^2} \right]_0^b$$

$$= \frac{16a^2b}{3}$$

①

①

①



$$\delta V \doteq 2\pi r h \delta x$$

$$= 2\pi (4-x) \left(\frac{x^2}{x^2+1} \right)$$

$$= 2\pi (4-x) \left(\frac{x^2+1-1}{x^2+1} \right)$$

$$= 2\pi (4-x) \left(1 - \frac{1}{1+x^2} \right)$$

①

①

①

(ii) $V = \int_0^2 2\pi (4-x) \left(1 - \frac{1}{1+x^2} \right) dx$

$$= 2\pi \int_0^2 \left(4 - \frac{4}{1+x^2} - x + \frac{x}{1+x^2} \right) dx$$

$$= 2\pi \left[4x - 4 \tan^{-1} x - \frac{x^2}{2} + \frac{1}{2} \log(1+x^2) \right]_0^2$$

$$= 2\pi \left(6 - 4 \tan^{-1} 2 + \frac{1}{2} \log 5 \right)$$

$$7) (a) (i) \quad m \ddot{x} = -mk(1+v^2)$$

$$\ddot{x} = -k(1+v^2)$$

①

$$\frac{dv}{dt} = -k(1+v^2)$$

$$\frac{dt}{dv} = -\frac{1}{k} \cdot \frac{1}{1+v^2}$$

①

$$\int_0^T dt = -\frac{1}{k} \int_0^0 \frac{dv}{1+v^2}$$

$$T = \frac{1}{k} [\tan^{-1} v]_0^0$$

①

$$= \frac{1}{k} \tan^{-1} 0$$

$$(ii) \quad v \frac{dv}{dx} = -k(1+v^2)$$

①

$$\frac{dv}{dx} = -k \cdot \frac{1+v^2}{v}$$

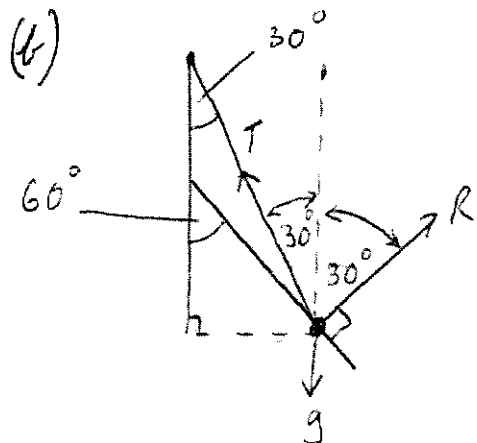
$$\frac{dx}{dv} = -\frac{1}{k} \cdot \frac{v}{1+v^2}$$

$$\int_0^x dx = -\frac{1}{k} \int_0^0 \frac{v dv}{1+v^2}$$

$$X = +\frac{1}{2k} [\log(1+v^2)]_0^0$$

$$= \frac{1}{2k} \log(1+v^2)$$

①



① Forces

① Angles

7 ctd

Horizontal: $T \sin 30^\circ - R \sin 30^\circ = \frac{mv^2}{r}$ $m=1, r=0.5$ ①

$$T - R = 4v^2 \text{ --- ①}$$

Vertical: $T \cos 30^\circ + R \cos 30^\circ = mg$ ①

$$T + R = \frac{2g}{\sqrt{3}} \text{ --- ②}$$

(ii) $v=1 \therefore T - R = 4$ --- ③

② + ③

$$2T = \frac{2g}{\sqrt{3}} + 4$$

$$T = \frac{g}{\sqrt{3}} + 2, \quad g = 9.8$$

$$\approx 7.66 \quad (2 \text{ decimal places}) \quad ①$$

sub in ③

$$R \approx 3.66 \quad (2 \text{ decimal places}) \quad ①$$

(iv) $T_{\max}$ when $R=0$

$\therefore T = \frac{2g}{\sqrt{3}}$ is the upper limit for T

① $R=0$ or $R>0$

① Answer

(or $R = \frac{2g}{\sqrt{3}} - T$ $R > 0 \therefore \frac{2g}{\sqrt{3}} - T > 0$
 $T < \frac{2g}{\sqrt{3}}$)

(v) $R=0 \Rightarrow 4v^2 = T = \frac{2g}{\sqrt{3}}$

$$v^2 = \frac{g}{2\sqrt{3}}$$

① v^2

$$v = \left(\frac{g}{2\sqrt{3}} \right)^{\frac{1}{2}}$$

① Answer.

or $R = T - 4v^2 > 0$

$$v^2 < \frac{T}{4}, \quad v < \left(\frac{g}{2\sqrt{3}} \right)^{\frac{1}{2}}$$

$$(8) (a) (i) \hat{C}EF + \hat{C}FD = 90^\circ + 90^\circ \quad (\text{Given})$$

$$= 180^\circ$$

$\therefore DECF$ cyclic (opp $\angle$ s supplementary)

①

$$\hat{D}FB = \hat{D}GB = 90^\circ \quad (\text{Given})$$

$\therefore DFGB$ cyclic (DB diameter, $\hat{D}FB, \hat{D}GB$ $\angle$ s in semicircle) ①

$$(ii) \text{ Let } \hat{E}CD = x$$

$\therefore \hat{E}FD = x$ ($\angle$ s in same segment of circle ECFD)

But $\hat{G}BD = \hat{E}FD$ (ext $\angle$ of cyclic quad. DFGB = opp. int. $\angle$)

$$= x$$

$\therefore \hat{D}FG = 180^\circ - x$ (opp. supplementary $\angle$ s of cyclic quad DFGB)

$$\therefore \hat{E}FD + \hat{D}FG = x + 180^\circ - x$$

$$= 180^\circ$$

$\therefore E, F, G$ collinear

① 2 appropriate facts with reasons.

① Using 3 cyclic quads

① correct answer.

$$(b) (i) \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \tan x \, dx$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\sin x}{\cos x} \, dx$$

$$= -[\log(\cos x)]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \log \sqrt{2}$$

①

①

8) ctd

$$\begin{aligned}
 \text{(ii)} \quad I_n &= \int_0^{\frac{\pi}{4}} \tan^n x \, dx \\
 &= \int_0^{\frac{\pi}{4}} \tan^{n-2} x (\sec^2 x - 1) \, dx \quad (1) \\
 &= \int_0^{\frac{\pi}{4}} \tan^{n-2} x \cdot \sec^2 x \, dx - \int_0^{\frac{\pi}{4}} \tan^{n-2} x \, dx \\
 &= \frac{1}{n-1} \left[\tan^{n-1} x \right]_0^{\frac{\pi}{4}} - I_{n-2} \quad (2) \quad 1 \text{ each.} \\
 &= \frac{1}{n-1} - I_{n-2}
 \end{aligned}$$

iii) $I_5 = \frac{1}{4} - I_3$

$$\begin{aligned}
 &= \frac{1}{4} - \left(\frac{1}{2} - I_1 \right) \\
 &= -\frac{1}{4} + \log \sqrt{2}
 \end{aligned}$$

(c) If $(x-b)$ and $(x-c)$ are factors
 then $(x-b)(x-c)$ is a factor (1)
 Note: $(x-b)(x-c) = x^2 - (b+c)x + bc$

$$\begin{aligned}
 P(b) &= b^m (c^n - b^n) + b^m (b^n - c^n) + c^m (b^n - b^n) \\
 &= 0 \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 P(c) &= c^m (b^n - c^n) + b^m (c^n - c^n) + c^m (c^n - b^n) \\
 &= 0 \quad (1)
 \end{aligned}$$

$\therefore (x-b), (x-c)$ are factors.